

DiffAudit: Evidence-Contracted Membership Leakage Auditing for Diffusion Models

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Abstract—Membership inference against diffusion models is often reported as a scalar score, yet a deployable privacy audit needs more than a high AUC. The score must be tied to a fixed target model, a member/nonmember split, row-level score or response coverage, reproducible metrics, provenance, and a clear consumer boundary. We present DiffAudit, an evidence-contracted audit methodology for diffusion-model membership leakage. DiffAudit separates admitted evidence from research candidates and watch items, reports finite-tail low-FPR metrics with their packet denominators, and rejects attractive results whose artifacts do not support the claimed audit semantics. On the current evidence bundle, DiffAudit admits five cross-permission rows spanning black-box reconstruction risk, gray-box PIA, a stochastic-dropout comparator, white-box GSA, and a DPDM defense comparator. We also report a controlled H2 response-cloud case study: an output-output geometry scorer reaches AUC 0.9615 on an existing response cache and survives label-shuffle, seed-offset control, seed-stability, and cross-cache transfer checks, but fails to cleanly port to an SD/CelebA img2img contract. Finally, bounded scouts on ReDiffuse STL-10, CommonCanvas, MIDST, and feature-packet artifacts show why second-asset claims require artifact contracts rather than metric prose. DiffAudit is therefore a measurement framework: it exposes verified leakage where evidence supports it and prevents unverified portability claims where the public artifact surface is incomplete.

I. INTRODUCTION

Diffusion models can memorize or overfit training examples in ways that enable membership inference attacks. The literature has produced strong black-box, gray-box, and white-box signals, including reconstruction similarity, posterior-error statistics, proximal initialization, conditional likelihood discrepancy, reconstruction-based black-box attacks, and gradient-based attacks [1], [2], [3], [4], [5]. However, paper-level scores are not automatically audit evidence. A model owner, regulator, or downstream runtime needs to know which target checkpoint was audited, what row IDs define members and nonmembers, whether scores exist per sample, how low-FPR tails should be interpreted, and which claims are blocked by missing artifacts.

DiffAudit is built around this distinction. We treat a membership score as admissible only when it is connected to an evidence contract. The contract records target identity, split semantics, score or response coverage, metric provenance, artifact provenance, and the surface delta needed for portability claims. This paper argues that such evidence contracts are not paperwork; they are scientific controls. They prevent source-label confounding, prompt-only shortcuts, finite-tail overclaiming, and the common failure mode where a high-

level metric cannot be replayed or consumed by a real audit system.

Our current evidence supports three findings. First, a small admitted bundle can already demonstrate meaningful diffusion privacy risk across access levels: black-box reconstruction risk reaches AUC 0.837, gray-box PIA reaches AUC 0.841, and white-box GSA reaches AUC 0.998 under their stated contracts. Second, a new H2 response-cloud analysis suggests that repeated output geometry contains membership signal beyond simple seed-to-output distance. This signal is strong and controlled inside the H2 response-strength setting, but it remains a research candidate because the SD/CelebA img2img portability check is weak or not distinct from simpler distance features. Third, many plausible second-asset routes are scientifically useful precisely because they fail: ReDiffuse STL-10 has a clean split and scoreable short target but random-level metrics; CommonCanvas has a real response packet but weak scorers; Tracing the Roots has positive feature tensors but no raw image-level consumer contract.

The paper therefore makes the following contributions:

- We define an evidence-contracted methodology for diffusion membership auditing that separates admitted, candidate, support-only, watch, and negative evidence.
- We report a five-row admitted evidence bundle with unified AUC, ASR, TPR@1%FPR, TPR@0.1%FPR, cost, provenance, and boundary language.
- We introduce and evaluate an output-cloud geometry case study that survives several bounded controls while remaining honestly limited to its current response-contract family.
- We document negative and support evidence from public artifacts, showing how missing checkpoints, broken row bindings, weak transferred signals, and consumer-boundary mismatches block overbroad portability claims.

II. RELATED WORK

A. Membership Inference and Evidence

Membership inference was introduced for discriminative machine-learning services as a way to infer whether a record was used during training [6]. Subsequent work connected the risk to overfitting and clarified the statistical interpretation of membership signals [7], [8]. Recent position work further argues that membership inference should not be treated as standalone proof that a model was trained on a particular datum [9]. DiffAudit adopts this caution as an audit-design principle: a high score is useful only after its target, split, metric, and consumer boundary are explicit.

TABLE I
DIFFAUDIT EVIDENCE-CONTRACT GATES.

Gate	Question answered
Target identity	Which model/checkpoint is being audited?
Split semantics	Which rows are members and nonmembers?
Score/response coverage	Is there row-level evidence for every sample?
Metric provenance	Can headline metrics be traced to rows?
Artifact provenance	Are artifacts hashable or repository-bound?
Consumer boundary	Who may consume the result and as what claim?

B. Diffusion Memorization and MIA

Diffusion models expose privacy risk both through memorized generations and through membership signals. Extraction and replication studies show that training examples or near-duplicates can appear in model outputs under some conditions [10], [11]. Diffusion MIA work then studies more direct member/nonmember decisions under different access levels. SecMI measures posterior-estimation error across denoising steps [1]; PIA reduces gray-box query cost through proximal initialization [2]; CLiD uses conditional likelihood discrepancy for text-to-image models [3]; ReDiffuse studies black-box variation-query attacks [12]; and GSA uses gradient statistics as a strong white-box signal [5]. DiffAudit does not replace these methods. It asks which method outputs can be safely compared, replayed, and consumed as audit evidence.

III. EVIDENCE CONTRACT

DiffAudit uses six gates before a result can support a portability or runtime audit claim. The target identity gate requires a fixed model architecture, checkpoint identity, training provenance, and accessibility. The split contract requires precise member and nonmember manifests. The score or response gate requires row-level coverage and labels. The metric gate requires AUC, ASR, $\text{TPR@1\%FPR}$, $\text{TPR@0.1\%FPR}$, nonmember denominator, and traceability to a score array or response packet. The provenance gate requires hashable or repository stored artifacts. The surface-delta gate checks whether a claimed second asset actually differs in model, data, modality, or response contract rather than repeating the same family.

Low-FPR metrics are reported as finite empirical packet readouts. For example, $\text{TPR@0.1\%FPR}$ on 100 nonmembers has a false-positive budget of 0.1 and should not be described as a calibrated continuous sub-percent operating point. This rule is important because many diffusion MIA packets are small, and strict-tail claims can otherwise sound stronger than the evidence permits.

IV. MEASUREMENT PROTOCOL

DiffAudit stores each result as one of five states. An admitted row can be reported to downstream consumers under its stated boundary. A candidate row is scientifically useful but not consumer-ready. A support-only row can motivate

or contextualize a mechanism but cannot serve as a direct benchmark. A watch item is tracked for future artifacts. A negative row answers a bounded decision question and usually closes a low-value expansion.

Metrics are recomputed from row-level scores or frozen response packets when available. We report AUC, attack success rate (ASR), $\text{TPR@1\%FPR}$, and $\text{TPR@0.1\%FPR}$, but we do not treat all four values equally. Low-FPR values are interpreted through the nonmember denominator: on a packet with 25 nonmembers, $\text{TPR@0.1\%FPR}$ has no continuous sub-percent meaning. This is why DiffAudit records both metrics and packet boundaries instead of publishing a leaderboard.

Table II summarizes how current non-admitted routes are used in the paper. The matrix is deliberately coarse: it is a claim-control device, not a new scoring system.

V. ARTIFACT CORPUS

The current paper corpus is intentionally small but heterogeneous. The admitted bundle contains five rows that already pass the consumer boundary. H2 output-cloud geometry contributes a mechanism candidate with strong internal controls. ReDiffuse STL-10, CommonCanvas, and MIDST contribute bounded negative results because each answers a specific route-selection question. Tracing the Roots contributes positive feature-packet evidence, while the collaborator Stable Diffusion ReDiffuse packet contributes a source-confounding stress test.

This corpus is not a broad survey of every diffusion MIA artifact. It is the evidence surface available under strict row-binding and no-overclaim rules. The next corpus expansion should be metadata-first: freeze inclusion criteria, check public target identity, inspect split and score/response availability, and only then decide whether any execution is justified. This keeps the measurement paper from turning artifact triage into uncontrolled engineering work.

VI. ADMITTED EVIDENCE BUNDLE

Table III summarizes the admitted rows. Each row is consumed with its cost and limitation language; none is treated as a universal benchmark.

VII. OUTPUT-CLOUD GEOMETRY CASE STUDY

The H2 response-strength cache contains repeated diffusion responses at timesteps 40, 80, 120, and 160. Instead of comparing a query image directly to outputs, the output-cloud scorer uses only geometry among outputs: within-step repeat RMSE, cross-timestep centroid RMSE, and small response-cloud PCA/Gram features. This design asks whether the response cloud itself carries membership information.

On the existing 512/512 cache, output-cloud logistic scoring reaches AUC 0.9615, ASR 0.9004, $\text{TPR@1\%FPR}$ 0.3340, and $\text{TPR@0.1\%FPR}$ 0.1172. It improves on raw H2 logistic by 0.0558 AUC and on lowpass H2 logistic by 0.0659 AUC. A label shuffle returns random-level AUC 0.5076. Because the historical cache used class-ordered seed offsets, DiffAudit released only a bounded shared-position control: with identical

Experiment or public artifact → target/split check → row-level score or response packet → metric replay → evidence state → consumer boundary

Admitted rows can flow to audit reports. Candidate, support-only, watch, and negative rows remain in Research unless a later review changes the contract.

Fig. 1. DiffAudit evidence-contract pipeline. The pipeline is intentionally claim-oriented: the same metric can become admitted evidence, a candidate case study, support-only context, or a bounded negative depending on provenance and consumer semantics.

TABLE II
CANDIDATE AND SUPPORT EVIDENCE GATE MATRIX. ‘PASS’ MEANS SUFFICIENT FOR THE STATED INTERNAL EVIDENCE ROLE, NOT AUTOMATIC PRODUCT ADMISSION.

Route	Target	Split	Evidence	Metric	Boundary	Paper use
Admitted bundle	Pass	Pass	Pass	Pass	Pass	Main audit rows
H2 output-cloud	Pass	Pass	Pass	Pass	Partial	Candidate mechanism
H2 img2img portability	Pass	Pass	Pass	Pass	Fail	Portability limit
ReDiffuse STL-10 scout	Pass	Pass	Pass	Pass	Fail	Bounded negative
CommonCanvas response packet	Partial	Pass	Pass	Pass	Fail	Bounded negative
Tracing Roots feature packet	Partial	Pass	Pass	Pass	Partial	Support-only
SD ReDiffuse collaborator	Partial	Partial	Pass	Pass	Fail	Source-confounded support

TABLE III
ADMITTED DIFFAUDIT EVIDENCE ROWS.

Track	Method	AUC	ASR	TPR@1%FPR	TPR@0.1%FPR	Role
Black-box	recon DDIM public-100 step30	0.837	0.740	0.220	0.110	controlled risk proof
Gray-box	PIA GPU512 baseline	0.841	0.786	0.059	0.012	baseline
Gray-box	PIA + stochastic dropout	0.828	0.768	0.053	0.010	provisional comparator
White-box	GSA 1k-3shadow	0.998	0.990	0.987	0.432	upper-bound attack
White-box	GSA against DPDM W-1	0.489	0.499	0.009	0.000	defense comparator

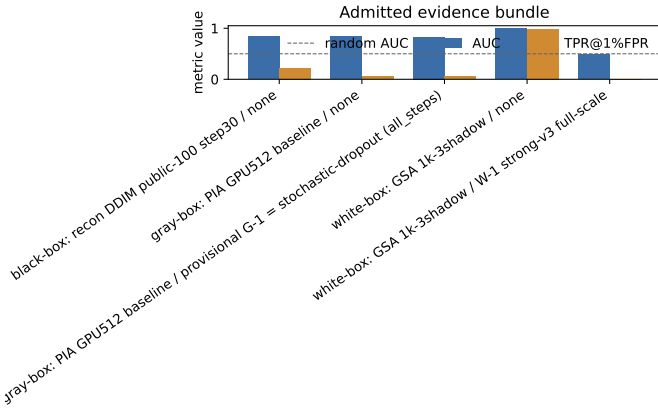


Fig. 2. AUC and TPR@1%FPR for admitted rows. The figure compares evidence under different access and defense contracts; it is not a single benchmark leaderboard.

per-position seed offsets for members and nonmembers, the scorer still reaches AUC 0.9678 at seed 176 and 0.9562 at seed 177. A fold-disjoint cross-cache transfer review gives AUC 0.9490 from seed 176 to 177 and AUC 0.9705 from seed 177 to 176.

The portability boundary is equally important. On the SD/CelebA img2img admission cache, output-cloud geometry reaches only AUC 0.7888 with zero strict tail recovery and underperforms the existing best simple-distance comparator by

0.088 AUC. A smaller stability cache is positive but still below simple distance. Thus the paper reports output-cloud geometry as a controlled mechanism candidate, not as an admitted black-box row.

VIII. NEGATIVE AND SUPPORT EVIDENCE

DiffAudit treats failed routes as measurement evidence when they answer a specific decision question. ReDiffuse STL-10 provides exact public split manifests and a locally scoreable bounded target, but fixed-timestep denoising-loss is random-level (AUC 0.4996) and a SimA-style score-norm observable remains weak (AUC 0.5053), even though ReDiffuse is a relevant black-box diffusion MIA reference point [12]. CommonCanvas/CopyMark has a real 50/50 response contract, yet conditional denoising-loss reaches only AUC 0.5148. Tracing the Roots feature tensors are positive (AUC 0.8158) [13], but they are not raw image-level evidence and lack target checkpoint/sample regeneration artifacts for the current consumer boundary. The collaborator Stable Diffusion ReDiffuse packet is replayable (AUC 0.7103), but source labels alone separate members and nonmembers perfectly, so it is a cross-source stress test rather than same-distribution membership evidence.

IX. DISCUSSION AND LIMITATIONS

The strongest criticism of DiffAudit is that evidence contracts may look like engineering governance rather than research. We argue the opposite: in diffusion membership

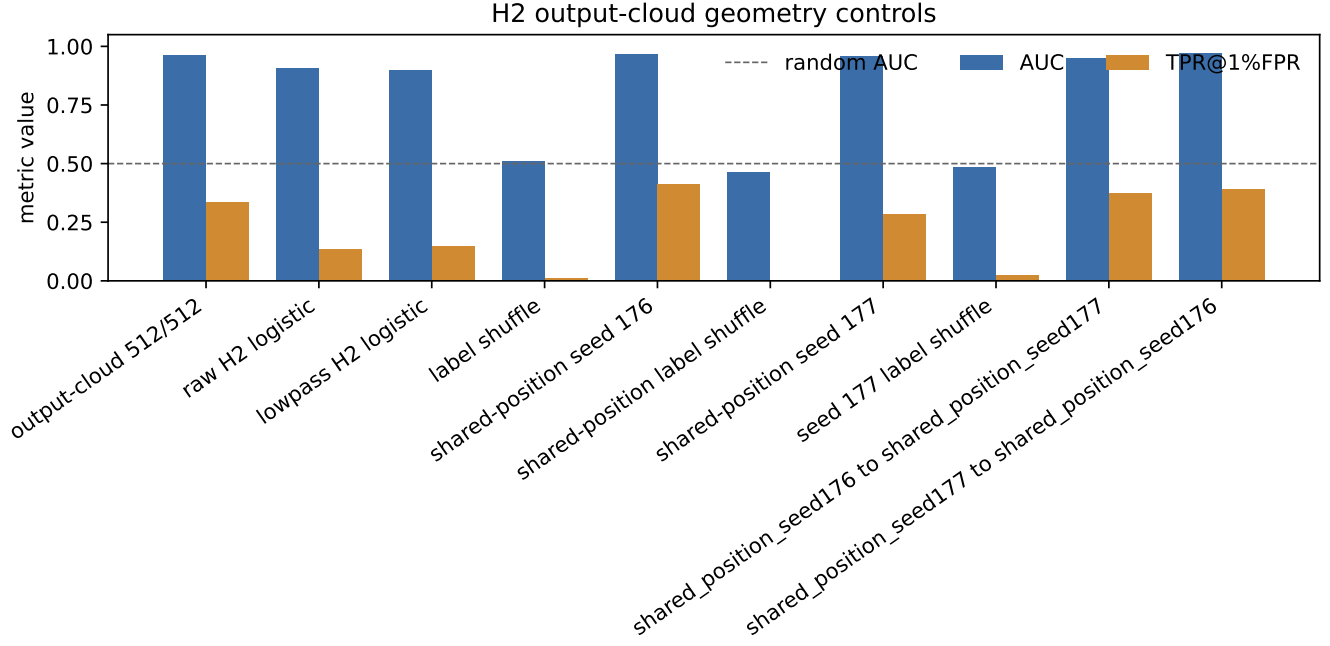


Fig. 3. H2 output-cloud geometry controls. The signal beats raw/lowpass baselines and survives label-shuffle, seed-offset control, seed stability, and cross-cache transfer. It remains a research candidate because portability to the SD/CelebA img2img contract is weak or not distinct from simple distance.

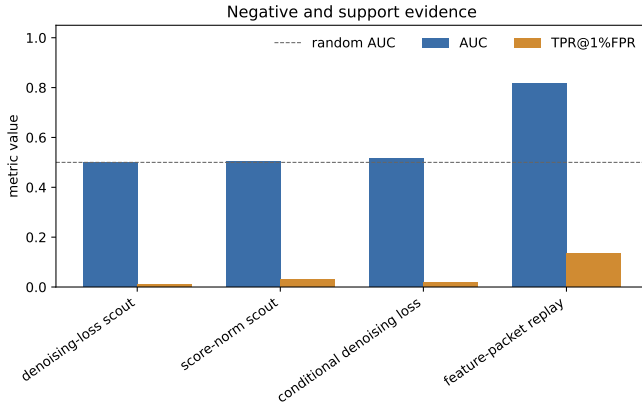


Fig. 4. Negative and support evidence. These results are not failures of the audit framework; they are examples of why scoreability, portability, and consumer semantics must be separated.

inference, evidence contracts are part of the measurement method. Without them, a high AUC can come from a source-label shortcut, a prompt-conditioned auxiliary feature, a non-replayable score array, or a finite-tail artifact. The H2 case study shows that DiffAudit can still discover new technical signals; the contract simply controls how far those signals are allowed to travel. This position is also consistent with broader caution that membership inference alone should not be overstated as definitive proof of training-data use [9].

This draft does not claim broad generalization across all diffusion models. It does not claim that H2 output-cloud

geometry is product-ready. It does not claim that bounded weak scouts disprove the original methods they resemble. The next scientific step for a full CCF-B+ submission is to add either a second checkpoint-bound response asset for the output-cloud mechanism or a larger, pre-registered public-artifact audit corpus for the evidence-contract methodology.

A. Threats to Validity

Internal validity. Some results are produced by local replay scripts and repository artifacts rather than by re-running complete upstream training pipelines. DiffAudit mitigates this by recording row-level provenance and by refusing to call bounded scouts full reproductions.

External validity. The admitted bundle is permission- and asset-bound. It should not be read as evidence that every diffusion model leaks at the same level, nor that one method dominates across modalities.

Metric validity. Low-FPR tails can be misleading on small packets. We therefore report them as finite empirical readouts and keep packet size and nonmember denominators attached to the claim.

Publication bias. Negative routes were selected because they affected DiffAudit route decisions. A standalone reproducibility paper would need a frozen corpus protocol before drawing broader conclusions about public artifact availability.

X. CONCLUSION

DiffAudit turns diffusion membership inference from isolated score reporting into evidence-contracted auditing. The current results show verified leakage under admitted black-box, gray-box, and white-box contracts; reveal a promising

output-cloud response geometry signal; and document why many public artifacts remain scientifically useful but non-admissible. This combination is the paper’s central point: publishable diffusion privacy evidence requires both strong signals and strict claim boundaries.

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